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Brief Report

Jupiter Mass Binary Objects Show a Minimum Acceleration

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Abstract

Forty-two Jupiter Mass Binary Objects (JuMBOs) have been discovered in the Trapezium Cluster: either brown dwarf stars or planets mutually orbiting in pairs. Here it is shown that, just as in galaxies and wide binaries, the mutual orbits of the objects in each of these twin systems deviate from the Newtonian and level off around a mutual acceleration of $2c^2/\Theta = 2 \times 10^{-10}$ m/s^2, supporting the minimum acceleration predicted by Quantised Inertia (QI), a theory that attributes inertial mass to an interaction between information horizons and quantum fields and predicts galaxy rotation without the need for dark matter. QI further predicts that the JuMBOs with separations of 400 AU should show orbital anomalies of 70 m/s. This could be tested using spectral Doppler data.

Keywords: Celestial mechanics; binaries; quantised inertia

1. Introduction

A team using the James Webb Space Telescope has discovered 42 pairs of Jupiter-Mass Binary Objects (JuMBOs) in the Trapezium open cluster of stars located in the Orion nebula [1]. Spectral information has shown that these JuMBOs are brown dwarfs or binary planets. It has been argued that some of these objects could be background galaxies reddened by the dust of the Orion nebula ([2]), but the Trapezium Cluster is out of the galactic plane (its galactic latitude is −19.4 degrees [3] and it is 390 pc from Earth [1] so contamination from background stars should be minimal [4] and it has been argued that it would be statistically impossible for background objects at random positions to mimic close binaries in this way [5]. These binary objects are not bound to stars, only to each other. Their separations range from 25 to 390 AUs and their masses from 0.7 to 13 times the mass of Jupiter [1].

Standard simulations cannot show any viable path to formation in the trapezium cluster [6], but a possible resolution lies in a relatively new theory for inertial mass called quantised inertia (QI) ([7,8]) suggested to address the problem of galaxy rotation. The model works as follows. When any object accelerates it sees so-called Unruh radiation ([9–11]) and an information horizon forms in the direction it is accelerating away from [12]. The existence of Unruh radiation was supported by [13] who interpreted radiation being emitted from plasmons propagating around gold nanotips as such, and more solidly by [14] who showed that emissions from positrons decelerating in a silicon crystal were consistent with it. In the quantised inertia model, the Unruh waves have to fit into the distance between the object and the horizon (nodes are only allowed at the horizon and object) so the Unruh radiation is damped by the horizon in the direction opposite to the acceleration and the Unruh radiation will be more intense in the other direction, pushing the object back against its initial acceleration. This predicts inertia ([8,15]). The model also predicts a

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slight modification to the usual inertial mass because some of the Unruh radiation will be suppressed isotropically, by the distant Hubble horizon. This will make this mechanism for inertia less efficient, reducing inertial mass in a new way for very low accelerations for which Unruh waves are very long ([7]). The overall model, called quantised inertia, modifies the standard inertial mass (m) as follows:

$$m_i = m\left(1 - \frac{2c^2}{|a|\Theta}\right) \quad (1)$$

where c is the speed of light in a vacuum, Θ is the co-moving diameter of the observable universe (which is taken to be 8.8×10^{26} m, following [16] and $|a|$ is the magnitude of the acceleration of the object relative to surrounding matter. Equation (1) predicts that for terrestrial accelerations (e.g.,: 9.8 m/s^2) the second term in the bracket is tiny and we just get standard inertia, but in environments where the mutual acceleration is on the order of 2×10^{-10} m/s^2, for example at the edges of galaxies, the second term becomes larger and the inertial mass decreases in a new way. This modification does not affect tests of the equivalence principle that use torsion balances since the predicted motion is independent of the mass.

This difference has a significant effect on very low acceleration systems: it enables them to orbit unexpectedly fast and remain unexpectedly bound. In this way QI solves the galaxy rotation problem ([17]), that MoND also predicts (MoND stands for Modified Newtonian Dynamics [18]) and QI explains the anomalous orbits of wide binaries ([19]), that MoND cannot predict. Thus, QI gets rid of the need for dark matter. QI has an advantage over MoND because it gets rid of MoND's adjustable parameter, a_0. QI predicts this parameter itself as $a_0 = 2c^2/\Theta$ and therefore it has more predictive power. QI predicts changes in behaviour as Θ reduces which have been observed, either in the early, smaller cosmos ([17]), or in engineered metal cavities ([20]).

Another difference from MoND is that QI has no external field effect. The inertial mass of two interacting objects depends only on their mutual acceleration, not what external acceleration is applied. The QI theory may not yet be mathematically complete, as pointed out by [21], who derived a modified form of Equation (1). The modification is strongest at intermediate accelerations of 1.2×10^{-9} m/s^2 and it would affect some of the results presented here, but has less of an effect on the minimum acceleration of 2×10^{-10} m/s^2.

2. Method

Quantised inertia predicts that mutual accelerations (a), even in deep space, have a minimum, given by

$$a = \frac{2c^2}{\Theta} \quad (2)$$

where c is the speed of light and Θ is the co-moving diameter of the cosmos (8.8×10^{26} m, [16]). In the theory this minimum arises to ensure that the acceleration-dependent Unruh waves seen by objects do not become larger than the cosmos, and become unobservable. According to QI then, the gravitational (or other) acceleration in any given system must be such that

$$\frac{GM}{r^2} > \frac{2c^2}{\Theta} \quad (3)$$

where G is the gravitational constant, M is the mass and r is the separation. This means that the mutual accelerations of bound systems should always be more than this, and this is true of galaxies, wide binaries, and in this paper we look at the Jupiter Mass Binary Objects. Data regarding the 42 JuMBOs was taken from Table 1 of [1]. It has been assumed that the uncertainties in their mass are 50% based on the error bars from Figure 4 in [1] and

that results in a 50% error in the acceleration. The aim is to use these very low acceleration systems to test for QI which predicts that none should have a mutual acceleration below $2c^2/\Theta = 2 \times 10^{-10}$ m/s^2.

3. Results

Figure 1a shows, on the x axis, the separation between the two objects in each of the JuMBOs in Astronomical Units (AUs) and on the y axis, their mutual acceleration (on a log axis) (Data from [1]). The black dots show the accelerations calculated gravitationally from $a = 2GM/r^2$ where M was taken as the average mass of the two JuMBOs. As expected, at greater separations their mutual accelerations reduce but the acceleration levels off around a minimum of 2×10^{-10} m/s^2 (the red line) as predicted by quantised inertia (solely from cosmological observables).

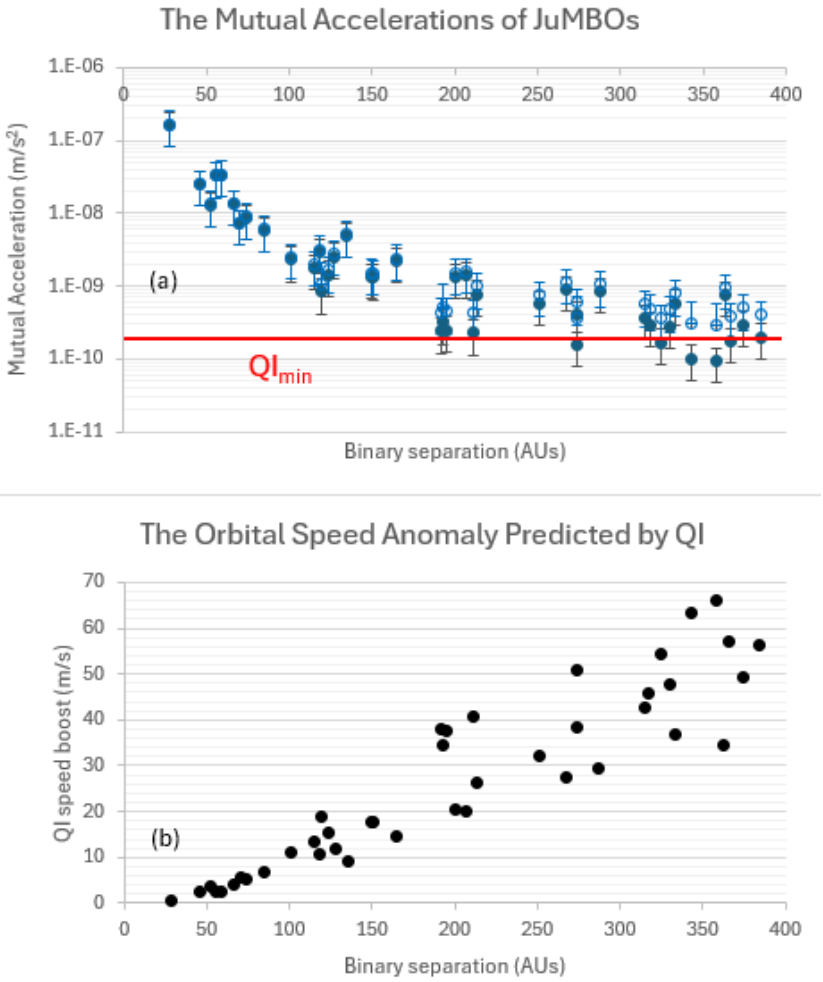


Figure 1. (**a**) The mutual accelerations predicted by Newtonian gravity (black) and by QI (blue), with error bars shown and (**b**) the excess orbital speed needed to keep mutual accelerations above the minimum predicted by QI.

Two of the pairs (JuMBOs 1 and 10) do have an acceleration that is marginally lower than the QI limit but see the discussion below. The plot shows a levelling off of the mutual acceleration around the QI minimum and in exactly the same way as for galaxy rotations and wide binaries. In all these cases the levelling begins when the accelerations approach 2×10^{-10} m/s^2 and dark matter cannot account for the anomalies on these smaller interstellar scales since it must stay spread out to explain the galactic cases, and

since the behaviour is numerically identical over such a wide range of scales, this implies that dark matter is not the solution.

However, QI predicts that the acceleration should be larger than that shown in Figure 1a. For these twin objects, in QI the balance is between gravity binding them and the centrifugal (inertial) force separating them, so

$$\frac{GMm}{r^2} = \frac{m_i v^2}{r} = m_i a \tag{4}$$

where m and m_i are the gravitational and inertial masses. Using Equation (1) for the inertia mass then we have a slight amendment, in that

$$\frac{GMm}{r^2} = m\left(1 - \frac{2c^2}{|a|\Theta}\right)a \tag{5}$$

$$a = \frac{GM}{r^2} + \frac{2c^2}{\Theta} \tag{6}$$

In other words, every acceleration should be augmented by $2c^2/\Theta = 2 \times 10^{-10}$ m/s^2 which will manifest as a faster mutual orbit than expected. When this is taken into account the actual dynamical acceleration ($a = v^2/r$) of the JuMBOs is predicted by QI to be larger, as shown by the blue dots on Figure 1a. All these values level off above the QI minimum ($2c^2/\Theta$), which is not surprising given the form of Equation (6), but it makes a testable prediction: the extra mutual acceleration of the binaries at the largest separations must be provided by an increase in the orbital speed of the objects in each pair. Normally we have

$$\frac{GM}{r^2} = \frac{v^2}{r} \tag{7}$$

so that $v = \sqrt{GM/r}$. In QI we have

$$\frac{GM}{r^2} + \frac{2c^2}{\Theta} = \frac{v^2}{r} \tag{8}$$

so the new, predicted orbital speed is

$$v = \sqrt{\frac{GM}{r} + \frac{2c^2 r}{\Theta}} \tag{9}$$

As shown in Figure 1b, as the separation increases from 50 towards 400 AU, the excess orbital speed predicted by QI increases from close to zero to almost 70 m/s. This orbital anomaly may be observable using telescopes such as HARPS (High Accuracy Radial Velocity Planetary Searcher, [22]) which has a velocity resolution of 1 m/s.

4. Conclusions

New data from the James Webb Space Telescope has identified 42 Jupiter Mass Binary Objects (JuMBOs) showing behaviour similar to that of stars in galaxies and wide binaries, in that the mutual accelerations of the objects in each binary level off above a minimum acceleration of $2c^2/\Theta = 2 \times 10^{-10}$ m/s^2.

As for wide binaries, this is evidence against dark matter, since this behaviour is identical to that seen in galaxies, but dark matter cannot be applied to both scales as it must be smooth on galactic scales to explain those systems.

The orbits of these JuMBOs can be explained simply by quantised inertia which predicts that their mutual accelerations must level off at the predicted minimum value

so that the Unruh wavelengths they see do not exceed the cosmic diameter and therefore become unobservable.

The alternative derivation of quantised inertia suggested by [21] would affect some of the results shown in Figure 1 and many more observations need to be collected to confirm that the JuMBOs are indeed binary objects.

QI predicts that the objects with 400 AU separations should show an anomaly in their mutual orbit of up to 70 m/s which could be detectable from the absorption spectra, given enough spatial and spectral resolution.

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